Karen E. Schroeder

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SUMMARY

My research is focused on the development of brain-computer interfaces, driven by a better understanding of how our brains plan and control movements. I am currently working on a novel locomotion BMI driven by rhythmic neural activity, with applications to wheelchair control and other prosthetic devices.

POSITIONS

Columbia University
Postdoc, Lab of Prof. Mark Churchland, Neuroscience Dept. 2016-present

EDUCATION

University of Michigan
Ph.D. Biomedical Engineering, Lab of Prof. Cynthia Chestek 2016

Duke University
B.S.E. Biomedical Engineering and Neuroscience 2010

WORKS IN PROGRESS

[1] **Schroeder KE***, Perkins SM*, Wang Q, Churchland MM, "A brain-machine interface for locomotion driven by subspace dynamics," *in preparation.*

[2] redacted

[3] redacted

PUBLICATIONS

[4] Bentley, J.N., Khalsa, S.S.S., Kobylarek, M., **Schroeder, K.E.,** et al. (2018) A simple, inexpensive method for subcortical stereotactic targeting in nonhuman primates. J. *Neuroscience Methods* 305.

[5] **Schroeder, K.E.,** Irwin, Z.T., Bullard, A.J., et al. (2017) Robust tactile sensory responses in finger area of primate motor cortex relevant to prosthetic control. *Journal of Neural Engineering* 14.

[6] Irwin, Z.T., **Schroeder, K.E.,** Vu P.P., et al. (2017) Neural control of finger movement via intracortical brain–machine interface. *Journal of Neural Engineering* 14.

[7] **Schroeder, K.E.,** and Chestek, C.A. (2016) Intracortical brain-machine interfaces advance sensorimotor neuroscience. *Frontiers in Neuroscience* 10.

[8] **Schroeder, K.E.,** Irwin, Z.T., Gaidica, M., et al. (2016) Disruption of corticocortical information transfer during ketamine anesthesia in the primate brain. *NeuroImage* 134.

[9] Irwin, Z.T., **Schroeder, K.E.,** Vu P.P., et al. (2016) Chronic recording of hand prosthesis control signals via a regenerative peripheral nerve interface in a rhesus macaque. *Journal of Neural Engineering* 13.

[10] Malaga, K.A., **Schroeder, K.E.**, et al. (2015) Data-driven model comparing the effects of glial scarring and interface interactions on chronic neural recordings in non-human primates. *Journal of Neural Engineering* 13.

[11] Irwin, Z.*, Thompson, D.*, **Schroeder, K.**, et al. (2015) Enabling Low-power, Multi-modal Neural Interfaces through a Common, Low-bandwidth Feature Space. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* PP, 1–1.

CONFERENCE PAPERS + PRESENTATIONS

[1] Ganem G, **Schroeder KE**, Perkins SM, Churchland MM, Cunningham J, "Deep random splines for point-process density estimation," *ICLR* 2019.

[2] **Schroeder KE**, Perkins SM, Wang Q, Churchland MM, "A brain-machine interface for locomotion driven by subspace dynamics," *COSYNE*, 2019

[3] **Schroeder KE**, Perkins SM, Wang Q, Churchland MM, "Virtual navigation via a closed-loop brain-machine interface," *SFN* 2018

[4] **Schroeder KE,** Cheng C, Zhang H, Wellner AW, Diester I, Kotov NA, Chestek CA, "Aramid nanofiber-epoxy coated intracortical LEDs for optogenetic stimulation," *Neural Interfaces Conference* 2014

[5] **Schroeder KE**, Snellings AE, Patel PR, Kipke, DR, "Biosensor arrays for concurrent subcortical recordings of neurotransmitters and electrophysiology in behaving rats," *SFN* 2012.

[6] Gibson MD, **Schroeder KE**, Kipke DR, "Neural Biosensor Arrays Provide Novel Interface for Simultaneous Recordings of Multiple Neurotransmitters and Electrophysiology," *SFN* 2011.

OTHER WRITINGS

- "How Fickle Neurons Encode Stable Behavior," Simons Foundation Collaboration on the Global Brain (SCGB) News, May 2019
- "Multitasking Neural Networks Balance Specialization With Flexibility," Simons Foundation Collaboration on the Global Brain (SCGB) News, March 2019 Online

INVITED TALKS

- Simons Collaboration on the Global Brain NY-area postdoc meeting 2019 Jan
- Cambridge University, Computational Neuroscience talk series 2018 Nov
- Columbia University, Zuckerman Institute Postdoc Seminar Series (ZIPS) 2018 Oct
- Backyard Brains, Ann Arbor MI 2016 May
- University of Michigan, Neural Engineering graduate course 2015 Apr

AWARDS

- COSYNE Travel Grant 2019
- Outstanding Research Mentor Award, U of M 2015
    - Only 6 awards given from pool of hundreds of faculty and grad student mentors
- Distinguished Leadership Award, UM College of Engineering 2015

- Awarded for contributions to gradute symposium
- Rackham International Research Travel Award 2015
- U.S. Dept. of Edu. Grad Assist. in Areas of National Need Fellowship 2013
- Rackham Merit Fellowship, U of M 2010
- NSF REU Fellowship, Biophotonics 2007

OTHER TRAINING

- MathWorks Research Summit 2019
- Grossman Workshop, Quantifying Structure in Large Neural Datasets 2017
- Primate Neuropixels Design Workshop, Janelia Research Campus 2017
- Methods in Computational Neuro Summer School, Wood's Hole MA 2016

LEADERSHIP + SCIENTIFIC SERVICE

- Blogger, Simons Collaboration on the Global Brain 2019-pres.
- Volunteer, Imagine Science Film Festival (ISFF) 2018-pres.
- Ad-hoc reviewer: Nature Comms, Journal Neural Eng 2017-pres.
- Co-chair, Engineering graduate symposium, UofM 2014
- Organizer, Translational Neural Engineering Journal Club 2013-2014
- Undergraduate research mentor 2014-2016

OUTREACH

- Volunteer, BioBus Harlem 2016-pres.
- Volunteer, Zuckerman Institute 'Saturday Science' 2016-pres.
- Volunteer, Q&A panels for grad/undergrad mentorship (UofM) 2014-2016
- Volunteer, Women in Science and Engineering (WISE) 2011-2014
    - Organized and ran various soldering and electronics workshops for middle and high-school aged girls, including portions of the GISE summer camp.
- Judge, Southeast Michigan Science Fair 2011-2016
- Teacher of 'Techtronics' grade 6-7 after-school science program 2008-2010

ENGINEERING PROJECTS

- Serial comm interface for moving acute electrode microdrives 2018
- Cheat-proof liquid reward nozzle for primate experiments 2017
- Custom microcontroller circuit for precise liquid reward dispensing 2017
- Multi-frequency haptic stimulation device for sensation experiments 2015
- Waterproof uLED array for cortical optogenetic stim of rodent ctx 2014

TEACHING

- Graduate Student Instructor: BME 241, Introductory Biomed Eng Laboratory

- Circuits and bioinstrumentation, materials testing, cell culture, and imaging

- Midterm student feedback: rated 4.92/5 in 'Overall, the instructor is an excellent teacher'

Students said: "Karen is great. She is very helpful." "The lab is very organized and well-run." "She is very helpful. I haven't seen anything I would change."

- Taught invited lectures for graduate-level Neural Engineering course, 2015